



Detection of Monkey pox through Machine Learning

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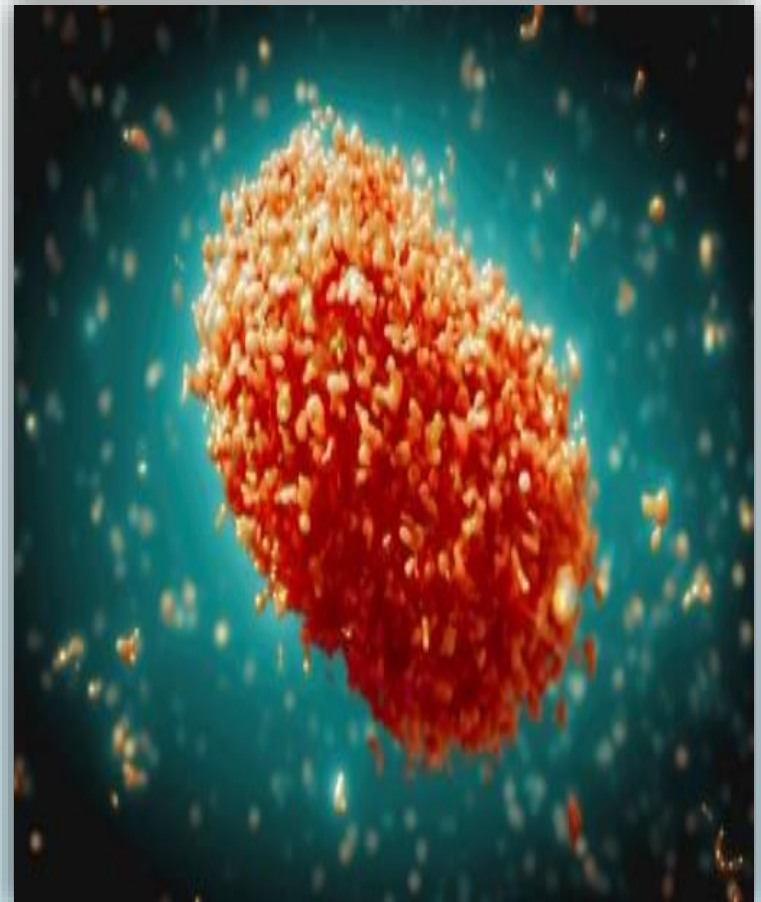
Introduction

Monkey Pox is a viral disease caused by the **Monkey Pox virus**, which belongs to the same family as the **smallpox virus**. It is a **zoonotic disease**, meaning it can spread from animals to humans, and also through human-to-human contact.

Monkey Pox typically present clinically with fever, rash and swollen lymph nodes and may lead to a range of medical complication.

Diagnosis done through histopathology, virus isolation and PCR test.

Occurs primarily in tropical rainforest areas of central and west Africa and is occasionally exported to other regions.



Transmission Method

Human-to-Human Transmission:

- Direct contact with skin lesions, rashes, or body fluids of an infected person.
- face-to-face contact (talking closely).
- Touching contaminated objects like clothing, bedding, or personal items.



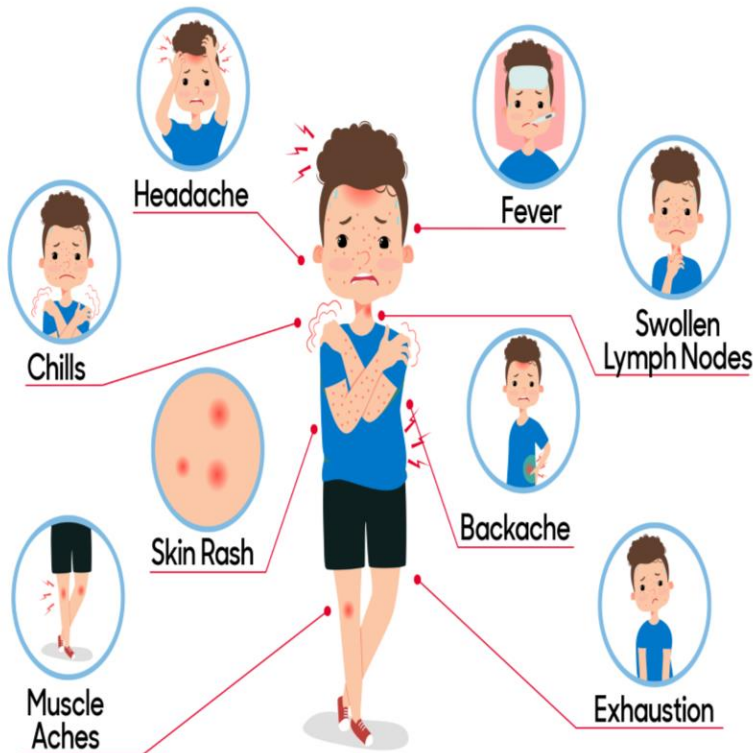
Animal-to-Human Transmission:

- Bites or scratches from infected animals (such as rodents or primates)
- Handling or consuming undercooked meat from infected animals.



Common Symptoms

Monkeypox Symptoms



1. Early Symptoms (Flu-like) – First Few Days:

Fever (one of the first signs)

Headache

Muscle aches (myalgia)

Swollen lymph nodes

2. Rash and Skin Lesions –(After 3 Days)

Starts as flat red spots → becomes fluid-filled blisters

then **scab over** Can appear **anywhere** (face, hands, feet, chest, genitals, etc.)

Cough

Nasal congestion

Eye irritation (if lesions spread near the eyes)

Prevention & Control

MONKEYPOX PREVENTION



AVOID TOUCH INFECTED
ANIMAL



AVOID TOUCH INFECTED
CLOTHES



WASH YOUR HANDS



WEARING A MASK



KEEP DISTANCE



EAT COOKED FOOD

1. Avoid Close Contact:

**With Infected Individuals
With Animals**

2. Personal Hygiene:

• **Wash hands soap and water** or use **alcohol-based sanitizer**.

3. Protective Measures:

• **Wear gloves and masks** when caring for an infected person.

4. Vaccination:

The JYNNEOS vaccine is recommended for high-risk individuals.

5. Isolation & Medical Care:

Self Isolation & Seek medical attention if symptoms appear, especially fever, rash, or swollen lymph nodes.

Problem Statement

Detection of Monkey pox Disease through its symptoms.

Solution:

Using Machine learning model for detecting Monkey Pox faster and more accurate.

Data Overview

Data Collection resource: from kaggle.com

Number of Records: 25,000 patients

Number of Features: 11 attributes
(Systemic Illness, Rectal Pain, HIV, Oral Lesions, etc.)

Target Variable: Monkey Pox
(Positive/Negative)

```
import pandas as pd
df= pd.read_csv("DATA.csv")
print(df.info())
```

```
[13] ✓ 0.2s

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 11 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Patient_ID                           25000 non-null  object
 1   Systemic Illness                     18784 non-null  object
 2   Rectal Pain                          25000 non-null  bool
 3   Sore Throat                         25000 non-null  bool
 4   Penile Oedema                       25000 non-null  bool
 5   Oral Lesions                        25000 non-null  bool
 6   Solitary Lesion                     25000 non-null  bool
 7   Swollen Tonsils                     25000 non-null  bool
 8   HIV Infection                       25000 non-null  bool
 9   Sexually Transmitted Infection      25000 non-null  bool
10  MonkeyPox                           25000 non-null  object
dtypes: bool(8), object(3)
memory usage: 781.4+ KB
None
```

```
df.head(5)
```

✓ 0.0s Python

	Patient_ID	Systemic Illness	Rectal Pain	Sore Throat	Penile Oedema	Oral Lesions	Solitary Lesion	Swollen Tonsils	HIV Infection	Sexually Transmitted Infection	MonkeyPox
0	P0	NaN	False	True	True	True	False	True	False	False	Negative
1	P1	Fever	True	False	True	True	False	False	True	False	Positive
2	P2	Fever	False	True	True	False	False	False	True	False	Positive
3	P3	NaN	True	False	False	False	True	True	True	False	Positive
4	P4	Swollen Lymph Nodes	True	True	True	False	False	True	True	False	Positive

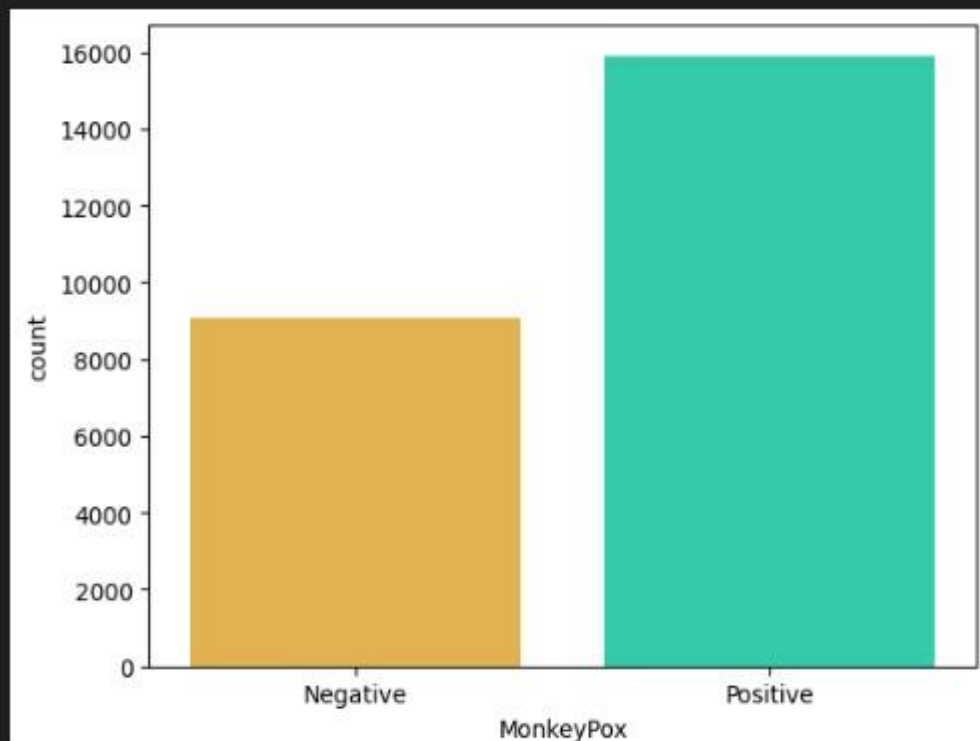
Monkey Pox case Distribution

```
print(df['MonkeyPox'].value_counts())  
sns.countplot(x=df["MonkeyPox"],palette='turbo_r')
```

✓ 0.4s

MonkeyPox
Positive 15909
Negative 9091
Name: count, dtype: int64

<Axes: xlabel='MonkeyPox', ylabel='count'>



In this Dataset Monkey Pox feature (column) has two class-

1. Monkey Pox(Positive)
patients counted: 15909

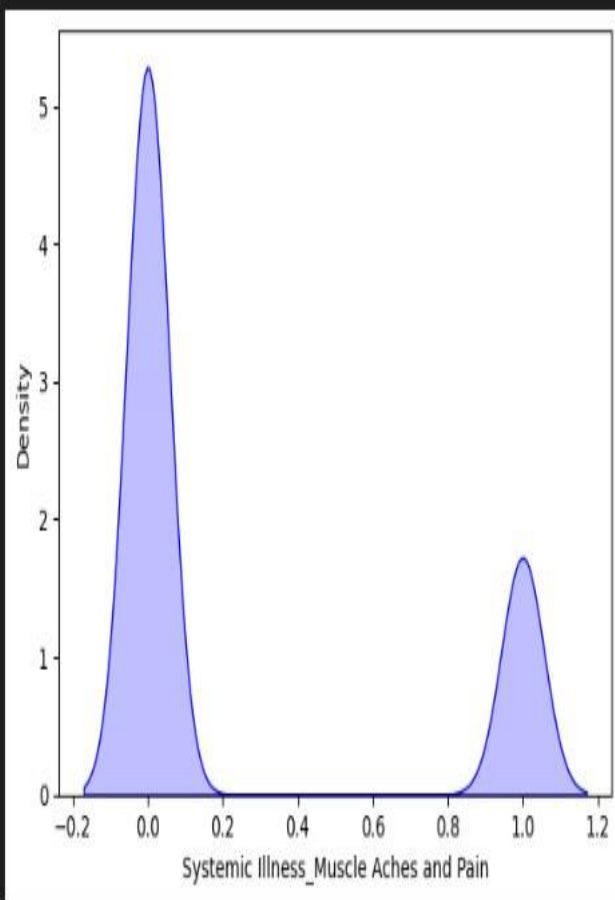
Class-2. Monkey Pox (Negative)
Patient counted: 9091



```
sns.kdeplot(x = 'Systemic Illness_Muscle Aches and Pain', data = df, color = 'blue', shade = True)
```

✓ 1.7s

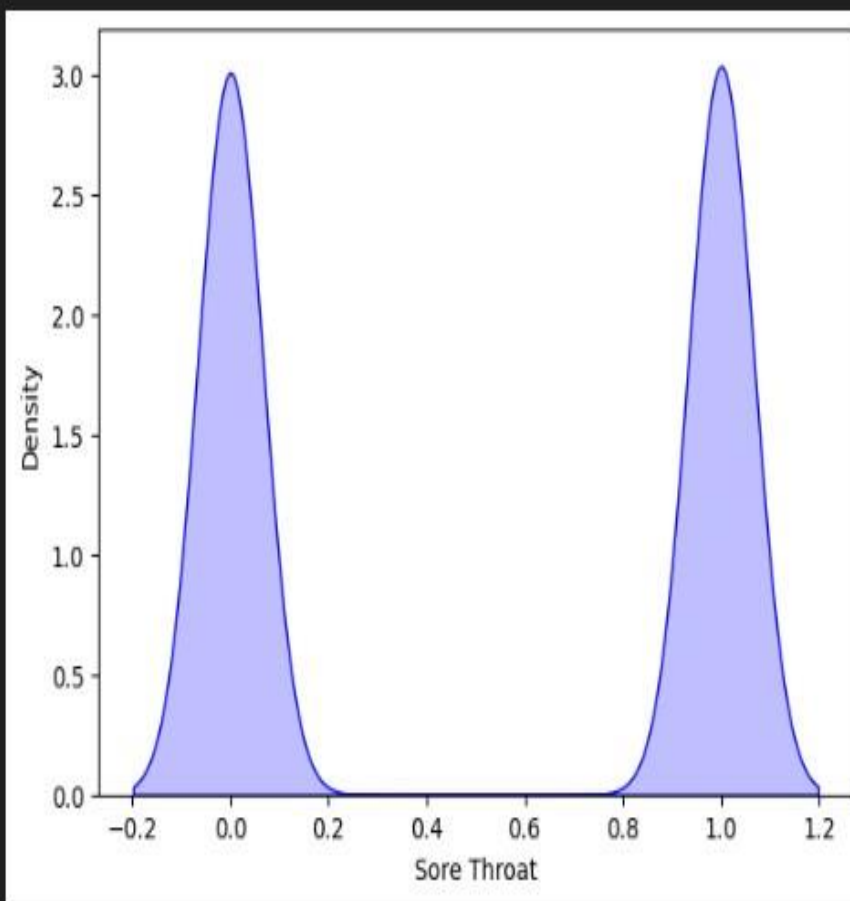
<Axes: xlabel='Systemic Illness_Muscle Aches and Pain', ylabel='Density'>



```
sns.kdeplot(x = 'Sore Throat', data = df, color = 'blue', shade = True)
```

✓ 0.9s

<Axes: xlabel='Sore Throat', ylabel='Density'>





```
sns.set_style("whitegrid")  
plt.figure(figsize=(8,5))  
sns.countplot(x='MonkeyPox',hue='Systemic Illness_Muscle Aches and Pain',data=df, palette='turbo_r')  
df["Systemic Illness_Muscle Aches and Pain"].value_counts()
```

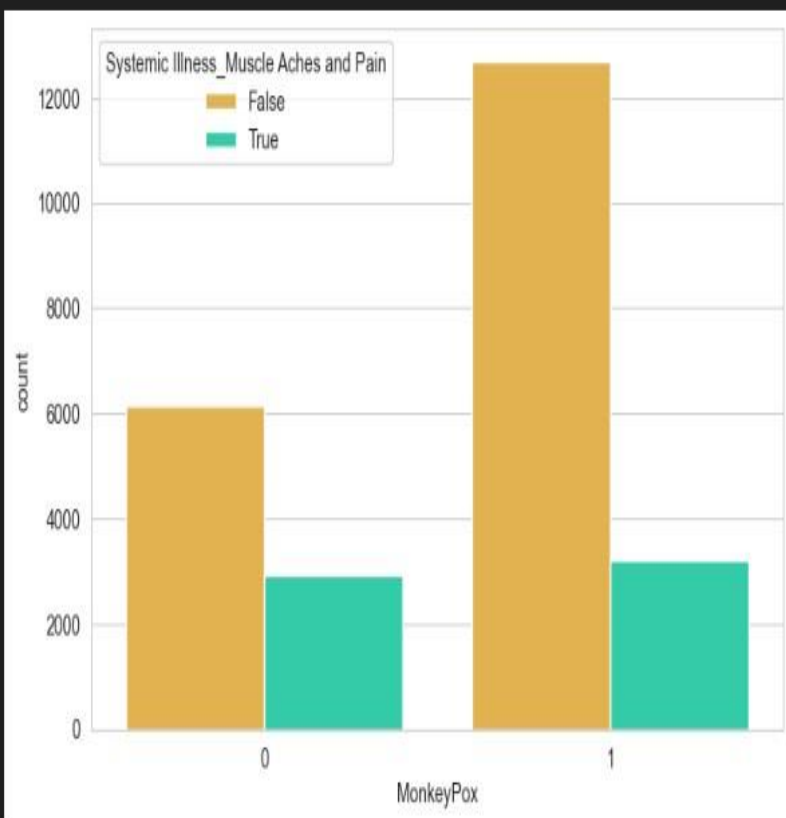
✓ 0.6s

Systemic Illness_Muscle Aches and Pain

False 18850

True 6150

Name: count, dtype: int64



```
plt.figure(figsize=(8,5))  
sns.countplot(x='MonkeyPox',hue='Sore Throat',data=df, palette='turbo_r')  
df["Sore Throat"].value_counts()
```

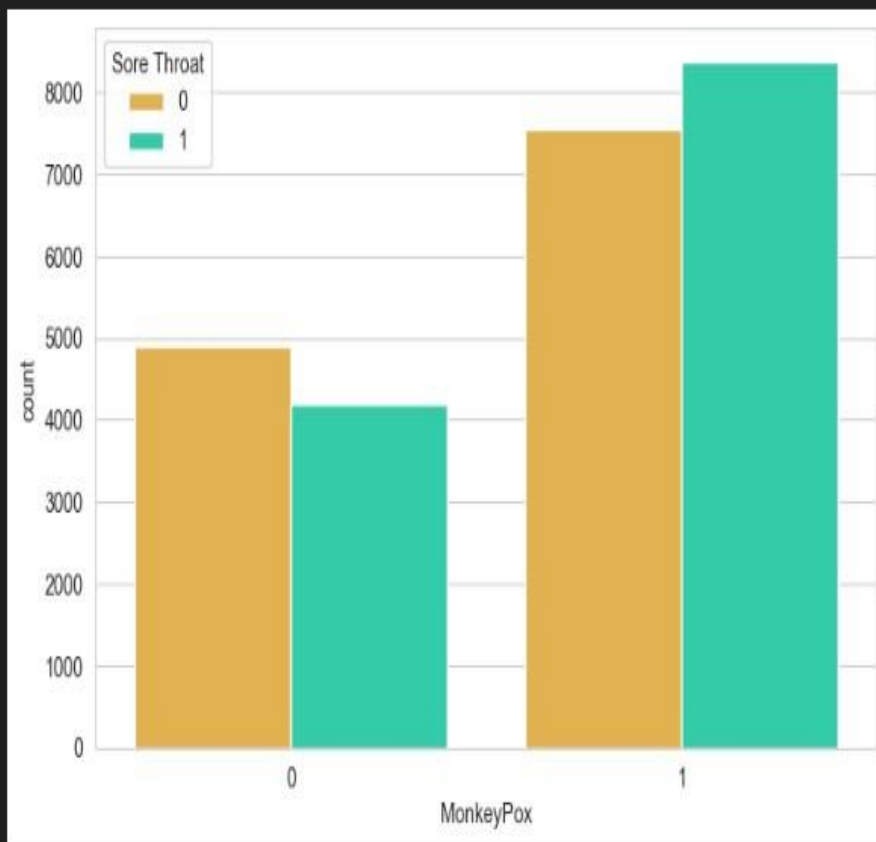
✓ 0.6s

Sore Throat

1 12554

0 12446

Name: count, dtype: int64



Decision Tree Algorithm For Prediction

		Actual Values	
		Positive	Negative
Predicted	Positive	TP=605	FP=1213
	Negative	FN=418	TN=2764

This Decision Tree model
Trained on training data
(20,000)
after that
I tested on testing data
(5000)
And
finally I got a prediction
how many patient
actually monkey Pox
diagnosed if monkey pox
symptoms is Positive or
Negative.

Logistic Regression Algorithm For Prediction

		Actual Values	
		Positive	Negative
Predicted	Positive	TP=672	FP=1146
	Negative	FN=487	TN=2695

This Decision Tree model
Trained on training data
(20,000)
after that
I tested on testing data
(5000)
And
finally I got a prediction
how many patient actually
monkey Pox diagnosed if
monkey pox symptoms is
Positive or Negative.

Overview

Step 1: Data Collection

- Obtained the Monkey Pox dataset containing patient symptoms and test results.

Step 2: Data Preprocessing

- Checked for missing values and handled them appropriately (e.g., imputation or removal).
- Encoded categorical variables using Label Encoding.
- Standardized numerical features where necessary.

Step 3: Exploratory Data Analysis (EDA)

- Visualized data distribution of key features.
- Analyzed correlations between symptoms and Monkey Pox positivity.
- Identified any outliers or inconsistencies.

Step 4: Feature Selection

- Used statistical tests and correlation analysis to select the most important features.
- Identified key variables: Systemic Illness, HIV Infection, Rectal Pain, STIs, and Oral Lesions.

```
Feature Importance Ranking:  
Systemic Illness      0.344819  
HIV Infection         0.122808  
Rectal Pain           0.119241  
Sexually Transmitted Infection 0.103493  
Solitary Lesion       0.064042  
Oral Lesions          0.062825  
Swollen Tonsils       0.062804  
Sore Throat           0.062338  
Penile Oedema         0.057630  
dtype: float64
```

Step 5: Model Selection & Training

•Applied Decision Tree Classifier:

- Trained a Decision Tree model to classify Monkey Pox-positive cases.
- Observed high accuracy but slight overfitting.

•Applied Logistic Regression:

- Trained a Logistic Regression model for better interpretability.
- Ensured good generalization and stable results.

Step 6: Model Evaluation

- Compared accuracy, precision, recall, and F1-score for both models.
- Decision Tree: High accuracy but prone to overfitting.
- Logistic Regression: More generalizable with good predictive power.

Step 7: Model Interpretation & Insights

- Decision Tree provided clear decision rules.
- Logistic Regression gave probability-based insights for each feature.

Conclusion

Prediction of Decision Tree Algorithm

Key predictors:

identified by the Decision Tree included Systemic Illness, HIV Infection, and Rectal Pain, indicating their strong influence on diagnosis.

The **Decision Tree model** achieved **high accuracy** on training data but suffered from **overfitting**, meaning it performed well on known data but lacked generalization on new data

It provided **clear decision rules**, making it useful for clinical decision-making and understanding the key symptoms leading to a **Monkey Pox-positive diagnosis**

- **True Positives (TP):** Correctly predicted Monkey Pox-positive cases(672).
- **False Positives (FP):** Incorrectly classified non-Monkey Pox cases as positive(1146).
- **False Negatives (FN):** Missed Monkey Pox cases, reducing recall(487).
- **True Negatives (TN):** Correctly identified non-Monkey Pox cases(2695)



```
cf_matrix=confusion_matrix(y_test,prediction)
print(cf_matrix)
```

✓ 0.0s

```
[[ 672 1146]
 [ 487 2695]]
```

Conclusion

Logistic Algorithm Prediction

- Provided a **stable and interpretable model** with good generalization.
- Showed **statistical significance of symptoms**, allowing for probability-based predictions.
- More suitable for understanding the **relationship between symptoms and MonkeyPox positivity**.

When to Choose

- If **interpretability** and **generalization** are needed, **Logistic Regression** is the better choice.
- If a **rule-based classification** is preferred, **Decision Tree** works well but needs pruning to avoid overfitting.



```
cf_matrix=confusion_matrix(y_test,prediction)
print(cf_matrix)
```

✓ 0.0s

```
[[ 672 1146]
 [ 487 2695]]
```